| | | | |
|---|---|---|---|
| 5 | **Charring test:**<br>Take a small amount of an organic compound in a dry test tube. Add 2 ml of conc $H_2SO_4$ to it, and heat the mixture. | Charring takes place with smell of burnt sugar | Presence of carbohydrate |
| | **Tests for Aliphatic or Aromatic nature:** | | |
| 6 | **Ignition test:**<br>Take small amount of the organic compound in a Nickel spatula and burn it in Bunsen flame. | (i) Burn with sooty flame<br>(ii) Burns with non sooty flame | (i) Presence of an aromatic compound<br>(ii) Presence of an aliphatic compound |
| | **Tests for an unsaturation:** | | |
| 7 | **Test with bromine water:**<br>Take small amount of the organic compound in a test tube add 2 ml of distilled water to dissolve it. To this solution add few drops of bromine water and shake it well. | (i) Orange - yellow colour of bromine water is decolourised<br>(ii) No Decolourisation takes place<br>(iii) Decolourisation with formation of white precipitate. | (i) Substance is unsaturated.<br>(ii) Substance is saturated.<br>(iii) Presence of an aromatic amine or phenol. |
| 8 | **Test with $KMnO_4$ solution:**<br>Take small amount of the organic compound in a test tube add 2 ml of distilled water to dissolve it. To this solution add few drops of very dilute alkaline $KmnO_4$ solution and shake it well. | (i) Pink colour of $KmnO_4$ solution is decolourised<br>(ii) No Decolourisation takes place | (i) Substance is unsaturated.<br>(ii) Substance is saturated. |
| | **TEST FOR SELECTED ORGANIC FUNCTIONAL GROUPS** | | |
| | **Test For Phenol** | | |
| 9 | **Neutral $FeCl_3$ test:**<br>Take 1 ml of neutral ferric chloride solution is taken in a dry clean test tube. Add 2 or 3 drops (or a pinch of solid) oforganic compound to it. If no colouration occurs add 3 or 4 drops of alcohol. | (i) Violet colouration is seen<br>(ii) violet – blue colouration is seen<br>(iii) green colouration is seen | (i) Presence of phenol.<br>(ii) Presence of α–naphthol<br>(iii) Presence of β– naphthol |



organic analysis.indd 257 19-01-2021 12:24:19